6 In the hairdryer shown in Fig. 6.1, an alternating current passes through the heating coil when the switch is closed.

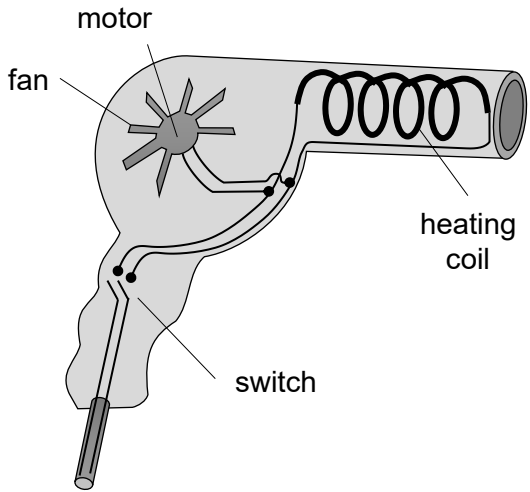

Fig. 6.1

(a) By reference to heating effect, explain what is meant by the root-mean-square (r.m.s) value of an alternating current.

.....

..... [1]

(b) The hair dryer is connected to the mains supply of 120 V r.m.s. and frequency 50 Hz. The heating coil delivers a power of 1200 W.

(i) Calculate the peak current through the heating coil.

peak current = ... A [2]

(ii) Write an equation, in terms of the elapsed time t, for the current that passes through the heating coil, given that the instantaneous power output is zero at $t = 0$ s. [1]

(c) The primary coil of a transformer is connected to a 2.4 kV r.m.s. supply. The secondary coil is connected to the hair dryer and the current flowing through the heating coil has the same value as that calculated in **(b)(i)**.

The transformer is non-ideal and electrical energy is converted to thermal energy in the transformer at a rate of 600 W.

Determine the r.m.s. current in the primary coil.

r.m.s. current = ... A [2]